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A General Paper on AI Based Mock Interview System

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ABSTRACT

Interviews play a very important role in the job selection process, but many candidates feel unprepared due to lack of practice, confidence, and proper feedback. Traditional interview preparation methods depend on mentors, friends, or self-practice, which are often inconsistent and limited. To address this problem, this paper proposes an Artificial Intelligence–based Mock Interview and Evaluation System [1], [2] that helps users prepare for real-life interviews in a structured and personalized manner.

The proposed system allows users to register, log in, and upload their resumes. The resume is analysed using Natural Language Processing techniques [1] to understand the user's education, skills, experience, and domain. Based on this analysis, the system conducts a mock interview consisting of three stages: HR interview, technical interview, and an aptitude test. The HR round focuses on communication skills and personality-based questions, the technical round evaluates subject knowledge related to the user's field, and the aptitude test assesses logical and problem-solving abilities.

User responses are evaluated using AI-based techniques such as semantic analysis and similarity scoring [1], [6]. After completing all stages, the system generates a detailed performance report highlighting strengths, weaknesses, and areas for improvement. This system provides a realistic interview environment, instant feedback, and repeated practice opportunities, helping candidates improve their confidence and interview performance. The proposed approach aims to make interview preparation more accessible, objective, and effective using Artificial Intelligence.

Keywords: Mock Interview System, AI-Based Interview Preparation, Resume Analysis, Performance Evaluation Report.

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INTRODUCTION

Interviews are an essential part of the recruitment process [1], as they help employers evaluate a candidate's knowledge, communication skills, confidence, and overall suitability for a job role. However, many candidates face difficulties during interviews due to lack of proper practice, nervousness, and limited access to structured feedback. Traditional interview preparation methods such as reading interview questions, practicing alone, or depending on friends and mentors are often inconsistent and do not provide objective evaluation. With the rapid growth of Artificial Intelligence (AI), interview preparation has evolved beyond manual methods. AI-based systems can simulate real interview environments and evaluate candidate responses [2], [5] in a more structured and unbiased manner. These systems are capable of analysing resumes, generating relevant questions, and assessing answers using techniques such as Natural Language Processing (NLP) and Machine Learning. This makes interview practice more personalized and effective for candidates from different domains.

The AI-Based Mock Interview and Evaluation System proposed in this paper is designed to help users prepare for real-life interviews by providing a complete interview simulation. The system begins with user registration and resume upload. The uploaded resume is analysed to understand the user's educational background, skills, and experience. Based on this information, the system conducts a mock interview consisting of three stages: an HR interview, a technical interview, and an aptitude test. Each stage focuses on evaluating different aspects of the candidate, such as communication ability, technical knowledge, and logical reasoning. Unlike traditional preparation methods, the proposed system provides instant and detailed feedback after the interview. It highlights the user's strengths and identifies areas that need improvement. This allows users to practice repeatedly and improve their performance over time. The main goal of this system is to create a realistic, intelligent, and accessible interview preparation platform that can help candidates build confidence and perform better in actual interviews.

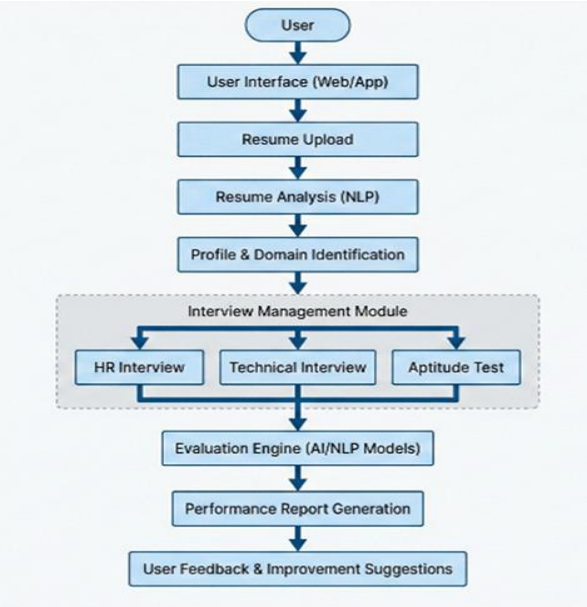
LITERATURE SURVEY

With the increasing competition in job markets, interview preparation has become an important area of research. Traditional mock interviews usually depend on trainers, faculty members, or peers, which makes the process time-consuming, subjective, and difficult to scale. To overcome these limitations, researchers have started using Artificial Intelligence to automate interview preparation and candidate evaluation [2], [1]. One major area of research focuses on resume analysis using AI. Many systems use Natural Language Processing techniques to extract information such as skills, education, and experience from resumes [1]. This extracted data is then used to understand the candidate's background and personalize interview questions. Studies show that automated resume analysis helps improve relevance and reduces manual effort, but challenges such as biased data, unstructured resumes, and lack of transparency still remain. Another important research direction is AI-based mock interview systems. These systems simulate real interview scenarios by asking HR and technical questions and evaluating user responses [2], [5]. Researchers have experimented with NLP models to analyse answer quality, clarity, and relevance. Some systems also use speech analysis to evaluate fluency and confidence [5]. While these platforms provide instant feedback and repeated practice opportunities, many of them struggle with domain-specific evaluation and deep understanding of technical explanations. Recent studies have also explored the use of computer vision and behavioural analysis during virtual interviews. Facial expressions, eye movement, head pose, and speech patterns are analysed [3], [6] to understand confidence, emotional state, and authenticity. Deep learning models have

shown good performance in detecting behavioural cues, but these approaches require large datasets and raise privacy concerns. Moreover, emotional analysis is not always accurate due to differences in individual behaviour and cultural factors researchers have further investigated emotional and psychological assessment using interview data. Although many of these studies are applied in healthcare or counselling domains, the underlying techniques for stress and emotion detection are useful for interview evaluation systems. These methods help identify nervousness, hesitation, and communication discomfort, which are important indicators during interviews.

From the existing literature, it is clear that AI-based interview systems provide advantages such as automation, personalization, and objective evaluation. However, current systems still face limitations related to domain adaptability, evaluation of soft skills, fairness, and data privacy[3], [6]. These gaps highlight the need for an improved AI-Based Mock Interview and Evaluation System that can provide realistic interview practice, meaningful feedback, and reliable performance analysis. The proposed system in this paper aims to address these challenges by combining resume analysis, adaptive questioning, and AI-driven evaluation techniques.

SYSTEM ARCHITECTURE



The AI-Based Mock Interview and Evaluation System follow a structured and sequential architecture designed to simulate a real interview environment while providing personalized evaluation and feedback. The system architecture clearly defines how data flows from the user to different processing modules and finally results in a performance report. The complete architecture is shown in the System Architecture Diagram provided above.

The process begins with the User, who interacts with the system through a User Interface (Web/App). This interface allows the user to register, log in, upload their resume, and participate in mock interviews. It acts as the main communication layer between the user and the system.

Once the user uploads the resume, it is passed to the Resume Upload Module, which securely stores the document and forwards it for analysis. The resume is then processed by the Resume Analysis Module (NLP). This module uses Natural Language Processing techniques to extract important information such as skills, education, experience, and technical keywords from the resume.

The extracted information is sent to the Profile and Domain Identification Module. This module determines the user's domain (for example, computer science, electronics, or mechanical engineering) and identifies the user's expertise level. This step is essential for selecting relevant interview questions and ensuring personalization.

After profile identification, control is transferred to the Interview Management Module, which is the core component of the system. This module manages and coordinates the entire mock interview process. It conducts three different interview stages:

- HR Interview, which focuses on communication skills and personality-based questions
- Technical Interview, which evaluates domain-specific knowledge
- Aptitude Test, which assesses logical reasoning and problem-solving ability

All user responses from these stages are collected and passed to the Evaluation Engine (AI/NLP Models). This engine analyses responses using AI-based techniques such as semantic similarity, keyword matching, and scoring algorithms. It evaluates factors such as correctness, clarity, relevance, and communication quality.

Based on the evaluation results, the system generates a detailed report through the Performance Report Generation Module. This report summarizes the user's performance across all interview stages and assigns appropriate scores.

Finally, the system provides User Feedback and Improvement Suggestions, highlighting strengths, identifying weaknesses, and suggesting areas for improvement. This feedback helps users understand their performance and prepare better for real-life interviews.

Overall, the architecture ensures a smooth flow of data, modular processing, and effective evaluation, making the system scalable, efficient, and suitable for intelligent interview preparation.

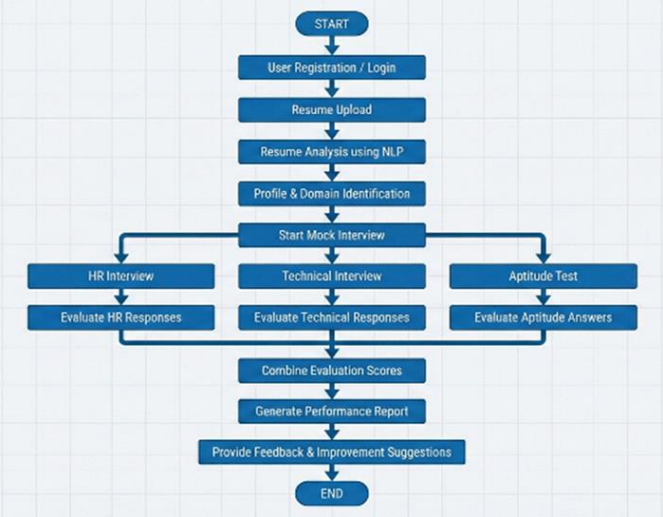
PROPOSSED SYSTEM

The proposed AI-Based Mock Interview and Evaluation System is designed to provide a complete, automated, and personalized interview preparation platform. Unlike traditional interview practice methods, this system uses Artificial Intelligence to analyse user profiles, conduct mock interviews, and generate meaningful performance feedback. The proposed system starts by allowing users to create an account and upload their resume. The resume is analysed using Natural Language Processing techniques to extract key information such as skills, education, experience, and domain-specific keywords. This information helps the system understand the user's background and customize the interview accordingly.

Based on the extracted profile, the system conducts a mock interview in three structured stages: **HR interview, technical interview, and aptitude test**. The HR interview focuses on personality-based and communication-related questions. The technical interview evaluates the user's subject knowledge according to their identified domain, such as computer science, electronics, or mechanical engineering. The aptitude test measures logical reasoning, analytical thinking, and basic problem-solving skills. During each stage, the system collects user responses in text or speech form. These responses are processed using AI-based evaluation techniques, including semantic analysis and similarity scoring. The system checks the relevance, correctness, clarity, and completeness of each answer rather than relying only on keyword matching. This helps achieve a more accurate and fair evaluation.

After all interview stages are completed, the system generates a detailed performance report. This report provides scores for each interview round and highlights strengths and weaknesses. In addition, the system suggests improvement areas, such as communication skills, technical concepts to revise, or reasoning ability. The proposed system aims to simulate a real interview environment while offering instant feedback and repeated practice opportunities. By using AI-driven evaluation and personalized questioning, the system helps users build confidence, improve interview skills, and perform better in actual job interviews.

FOW CHART



The flow chart represents the step-by-step operation of the AI-Based Mock Interview and Evaluation System, showing how the user interacts with the system and how data is processed at each stage.

The process begins with **user registration or login**, which ensures secure access to the platform. After successful login, the user **uploads their resume**, which serves as the primary input for personalizing the interview process.

Once the resume is uploaded, the system performs **resume analysis using Natural Language Processing (NLP)** techniques. Important information such as skills, education, experience, and domain-specific keywords is extracted from the resume. Based on this extracted data, the system performs **profile and domain identification**, which helps determine the type and difficulty level of interview questions.

After profiling, the system initiates the **mock interview process**, which is conducted in three stages. First, the **HR interview** evaluates the user's communication skills, confidence, and personality-based responses. Next, the **technical interview** assesses the user's subject knowledge according to their identified domain. Finally, the **aptitude test** measures logical reasoning and problem-solving abilities.

User responses from each stage are **evaluated individually** using AI-based evaluation techniques. After completing all interview stages, the system **combines the evaluation scores** to generate an overall performance assessment.

Finally, a **performance report is generated**, which highlights the user's strengths, weaknesses, and areas for improvement. The system then provides **feedback and improvement suggestions** to help the user prepare better for real-life interviews. The process ends after the feedback is presented to the user.

METHODOLOGY

The methodology describes how the AI-Based Mock Interview and Evaluation System is implemented to provide a structured and personalized interview preparation experience. The system follows a sequential and modular approach, ensuring smooth data flow, accurate evaluation, and meaningful feedback generation.

The process begins with user registration and authentication, where users create an account and securely log in to the platform. This step ensures that user data, interview attempts, and performance reports are stored and managed safely.

After login, the user is required to upload their resume. The uploaded resume is processed using Natural Language Processing techniques to extract relevant information such as educational background, technical skills, experience, and domain-specific keywords. This extracted data is cleaned and structured so that the system can accurately understand the user's profile.

Once the resume is analysed, the system performs profile creation and domain identification. Based on the detected domain and skill level, appropriate interview questions are selected. This personalization ensures that users receive questions relevant to their field and career background rather than generic interview questions.

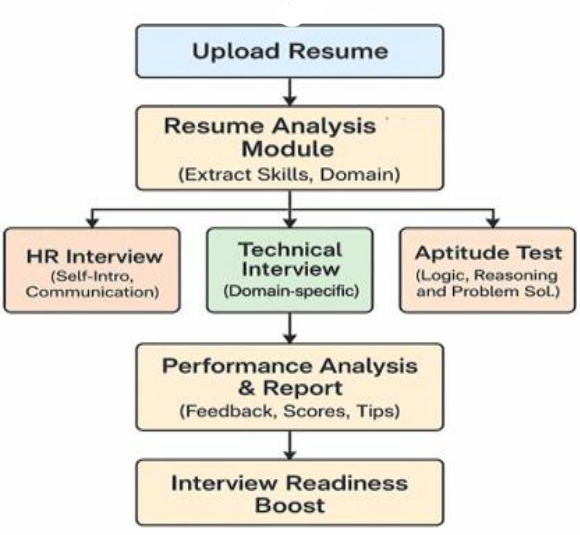
The mock interview is then conducted in three phases. The HR interview phase focuses on personality-based and communication-related questions to assess clarity of speech, confidence, and response structure. The

technical interview phase evaluates subject knowledge by asking domain-specific questions. The aptitude test phase measures logical reasoning, numerical ability, and analytical thinking.

During each phase, the user's responses are collected in text or voice format. These responses are processed using AI-based evaluation techniques such as semantic similarity analysis and scoring mechanisms. The system evaluates responses based on correctness, relevance, clarity, and completeness instead of relying only on keyword matching.

After completing all interview stages, the system aggregates the evaluation scores and generates a comprehensive performance report. This report highlights strengths, identifies weak areas, and provides improvement suggestions. The feedback enables users to understand their performance and improve through repeated practice.

Overall, the methodology ensures a realistic interview simulation, automated evaluation, and personalized feedback, making the system effective for interview preparation.



MATHEMATICAL MODEL OF THE SYSTEM

The mathematical model explains how the AI-Based Mock Interview and Evaluation System processes user inputs, evaluates responses, and generates performance results. The model uses basic set representation and scoring functions to describe the system behaviour.

A. System Representation

The complete system can be represented as:

$$S = \{I, P, O\}$$

Where:

- I represents the input set
- P represents the processing functions
- O represents the output set

B. Input Set

The input set consists of the user's resume and interview responses:

$$I = \{R, A\}$$

Where:

- R= User resume
- $A = \{a_1, a_2, a_3, \dots, a_n\}$ = User answers to interview questions

C. Resume Feature Extraction

The uploaded resume is converted into structured features using an embedding function:

$$F_R = E(R)$$

Where:

- $E(\cdot)$is an NLP-based embedding function (such as BERT)
- F_R represents extracted features like skills, education, and domain

These features are used to select appropriate interview questions.

D. Answer Evaluation Model

Each user response is converted into an embedding:

$$F_A = E(a_i)$$

The system compares the user's answer with a reference answer using cosine similarity:

$$Score_i = \cos(F_A, F_{ref})$$

Where:

- $F_{ref}$is the reference answer embedding
- $Score_i$represents the relevance and correctness of the answer

E. Overall Performance Score

The final performance score is calculated by averaging all individual question scores:

$$Final\ Score = \frac{1}{n}\sum_{i=1}^{n} Scor\ e_i$$

Where:

- $n$is the total number of interview questions

F. Output Set

The output of the system is represented as:

$$O = \{Final\ Score, Feedback\}$$

Where:

- **Final Score** indicates overall interview performance
- **Feedback** includes strengths, weaknesses, and improvement suggestions

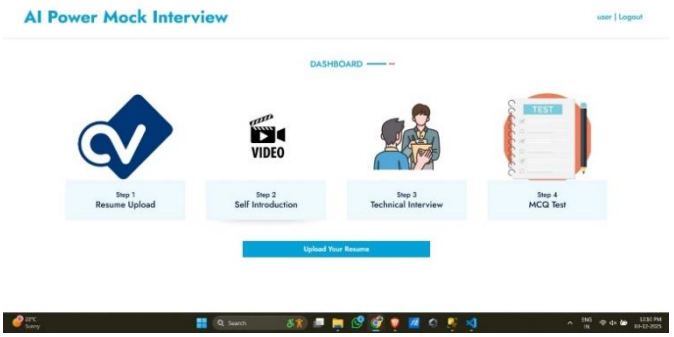
Summary

This mathematical model provides a simplified representation of how the system analyses resumes, evaluates interview responses, and generates performance results. It ensures objective scoring, consistency, and transparency in the interview evaluation process

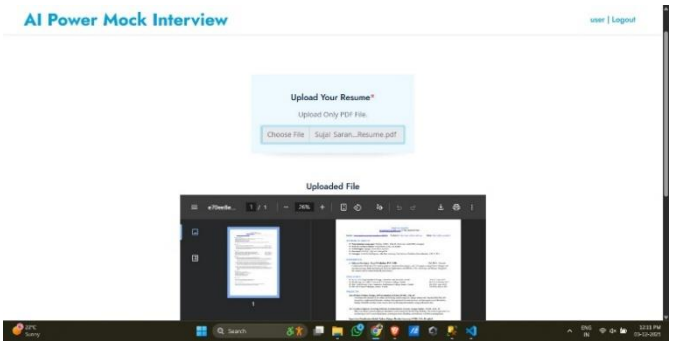
RESULTS



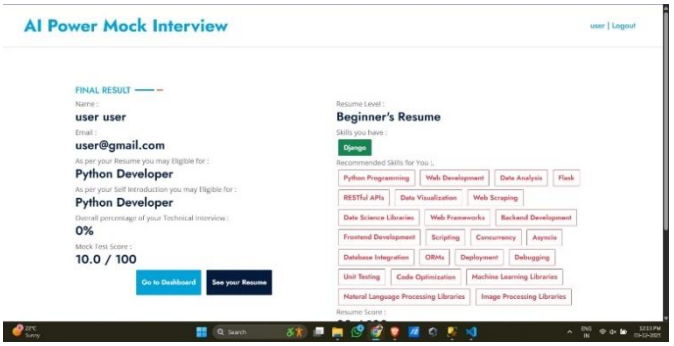
This image shows the home page of the "AI Power Mock Interview" web application. It provides a simple user interface with options for Login, Register, and Admin Login. The page introduces the platform, which is designed to help users prepare for interviews using AI-based mock interviews and evaluation features.



This image shows the dashboard of the AI Power Mock Interview system, where the interview process is divided into four steps: Resume Upload, Self Introduction, Technical Interview, and MCQ Test. It guides the user through each stage in a clear, step-by-step manner to complete the mock interview.



This image shows the resume upload interface of the AI Power Mock Interview system, where the user uploads a PDF resume. The uploaded resume is displayed for verification before being analysed by the system for interview personalization.



This image shows the final result page of the AI Power Mock Interview system, displaying the user's resume level, eligible interview domain, technical interview score, MCQ score, and recommended skills based on resume analysis and interview performance.

CONCLUSION

The AI-Based Mock Interview and Evaluation System provide an effective solution for interview preparation by simulating real interview scenarios using Artificial Intelligence. The system successfully analyses user resumes, identifies relevant domains, and conducts structured mock interviews including HR, technical, and aptitude rounds. By using AI-based evaluation techniques, the system offers objective assessment and instant feedback to users.

The results demonstrate that the proposed system can personalize interview questions, evaluate responses accurately, and generate meaningful performance reports. This helps users understand their strengths and weaknesses and improve their interview skills through repeated practice. Overall, the system reduces dependency on manual interview preparation methods and makes interview training more accessible, consistent, and efficient.

In the future, the system can be enhanced by incorporating advanced speech analysis, emotion detection, and real-time feedback mechanisms to further improve evaluation accuracy and user experience.

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